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VWAP Algorithm

Function

Executes a large order by participating proportionally in the market volume over a specified duration. The algorithm aims to achieve an execution price close to the market's Volume-Weighted Average Price (VWAP) while minimizing market impact.

Detailed Explanation

The Volume-Weighted Average Price (VWAP) algorithm dynamically adjusts its execution based on the market's traded volume. It ensures that the total quantity is executed over the specified duration while adhering to user-defined constraints such as participation rate and unfilled action.

1. Order Splitting:

- The algorithm calculates the order size for each execution interval based on the market's current volume ($V_{current}$):

$$Q_{interval} = V_{current} \cdot \frac{PR_{max}}{100}$$

Where PR_{max} is the maximum participation rate.

- Ensures that the total traded quantity does not exceed the remaining order size:

$$Q_{interval} = \min(Q_{interval}, Q_{remaining})$$

2. Dynamic Adjustments:

- Extend Execution Time:** If the unfilled action is 'extend', the algorithm recalculates the required execution duration:

$$T_{adjusted} = \frac{Q_{remaining}}{V_{current} \cdot \frac{PR_{max}}{100}}$$

- Increase Participation Rate:** If the unfilled action is 'increase', the algorithm adjusts the participation rate:

$$PR_{adjusted} = \frac{Q_{total}}{T_{total} \cdot V_{current}} \cdot 100$$

3. Monitoring:

- The algorithm continuously tracks:
 - Filled Quantity:**

$$Q_{filled} = \sum_{fills} Q_{fill}$$

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4. Completion.

- The algorithm terminates when the total traded quantity reaches Q_{total} or the time boundary is exceeded
- If $A_{unfilled} = fail$, the algorithm terminates with the remaining quantity unexecuted

Real-Life Example

Scenario: A trader wants to buy **500,000 shares** of XYZ stock over **2 hours**, targeting a maximum participation rate of **10%** of the market volume.

1. Inputs:

- $Q_{total} = 500,000$
- $T_{total} = 2 \text{ hours} = 7,200 \text{ seconds}$
- $PR_{max} = 10\%$

2. Execution Flow:

- The current market volume ($V_{current}$) is **50,000 shares per minute**
- **Interval Quantity:**

$$Q_{interval} = V_{current} \cdot \frac{PR_{max}}{100} = 50,000 \cdot \frac{10}{100} = 5,000 \text{ shares per minute}$$

- The algorithm places orders every second, proportional to the market volume

3. Adjustments:

- If the market volume drops significantly, the algorithm adjusts the execution time or participation rate based on the user-defined unfilled action

4. Outcome:

- The trader executes the order over 2 hours, aligning with the market volume and ensuring minimal market impact

Summary

The VWAP Algorithm is designed for:

- **Market-Driven Execution:** Executes orders proportionally to the market's traded volume
- **Dynamic Flexibility:** Adapts execution duration or participation rate based on the market and user preferences
- **Minimized Impact:** Aligns execution with market volume to achieve prices close to the VWAP

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